JUNIOR AI ENGINEER · AIE2609

Entity-level news sentiment for the S&P 500

Approach, key decisions, and what the results say

Son · 14 October 2026 · github.com/sonalain077/Whitehole_project

The task: a label per company, a score per ticker

100
news articles, 25 Aug to 28 Sep 2026

500
companies (503 rows, 3 dual classes merged)

233
article–company pairs in entities.csv

112
tickers scored in scores.csv

Score = Σ label × 0.5^(days / 7), label = +1 / 0 / −1, days measured to 29 Sep 2026 00:00 New York time.

The hard part: telling the company the news is about from companies merely mentioned: the bank issuing the rating, the media source, a peer used for comparison. Those are neutral.

Approach: the model reads, the code checks and computes

1

Load & check

CODE

Strict schemas, New York dates, invisible characters removed, GOOG → GOOGL

2

Classify

MODEL

One request per article with all 500 companies → company, role, label, exact quote

3

Validate & review

CODE + MODEL

Invalid answers re-asked with the reason; code flags missed names, the model re-reads only those articles

4

Vote

CODE

Steps 2–3 run three times independently; the majority answer per pair is kept

5

Score

CODE

Time-decay weights and sums → entities.csv, scores.csv, cost report

gpt-5.6-terra never computes a score, never sees other articles and has no tools. Every code step is covered by 100 offline tests with a fake model. One command: **python -m aie2609 run**.

Key decisions and trade-offs

Decision	Why	Evidence
Role before label	The PDF defines the rating bank, the source and a comparison as neutral	The schema forbids a non-neutral label for those roles
Validate everything, never default to neutral	LLM output is untrusted data	Exact-quote check; bounded retries that carry the reason
Code flags, model judges	Code spots names reliably, not implications	11–15 of 100 articles re-read per pass
Three passes + majority vote	The model varies on border cases	93.7 % pair agreement; accuracy 0.92 → 0.933
Full company list, no prefilter	A prefilter loses brands and subsidiaries	Waymo, CNBC, iShares, Xbox mapped to parents

Trade-off: three passes cost about 1.5× one pass (0.045 vs 0.030 USD per article) for fewer one-off errors and a confidence level per pair.

How it was built: two coding agents, one set of rules

RULES

AGENTS.md

Written from the PDF: only the two inputs, only gpt-5.6-terra, no manual edit of results, key and budget handling. Every batch ends with tests and a dated log entry.

BUILD · 12–13 OCT

Codex

Built the pipeline in small tested batches and produced the first results: the first entities.csv and scores.csv on the 100 articles.

REVIEW & FINISH · 14 OCT

Claude Code

Fresh context as an independent judge: audited Codex's work (no one-command run, results not reproducible), finished the pipeline, and judged the model's outputs to measure accuracy.

Budget, publication and the delivered version stayed my decisions. CLAUDE.md reuses AGENTS.md; docs/WORK_LOG.md keeps every command, cost and failure.

Reliability and cost, measured

0.933
estimated pair accuracy

93.7 %
pairs identical in 3 independent passes

$0.045
per article for 3 passes ($0.030 for one)

100
offline tests, no API needed

Version	Accuracy
Manual composition, prompt v4	0.878
One pass, prompt v5	0.907
One pass, prompt v6 (roles)	0.920–0.924
Three passes + vote (delivered)	0.933

All disagreements between versions plus 25 random checks, judged by Claude with the PDF definitions; I reviewed all 21 debatable cases and approved each one. Not human ground truth, and the judge also designed v6. The ranking still holds on the 20 clear-cut cases (19 / 17 / 9 correct).

Who stands out: Nike lowest, AMD highest

MOST NEGATIVE

NKE −1.39 · negative in all 4 articles over 3 weeks: BMO bearish, dropped from the S&P 100, "needs help", BofA downgrade to Underperform

NFLX −1.19 · two negative articles on 23 Sep: YouTube fights for creators, HSBC warning

BX −0.97 · one article on 28 Sep: the head of private equity leaves

BA −0.94 · one article on 28 Sep: 737 MAX software glitch under regulator review

TSLA −0.88 · most covered (9 articles): 3 positive in mid-September outweighed by 2 negative on 28 Sep

MOST POSITIVE

AMD +1.01 · $1T valuation and a BofA target raise outweigh a chip-sector slide on 13 Sep

UAL, DAL +0.92 each · Redburn stays bullish; both add capacity as Spirit is grounded

C +0.78 · own investment-banking growth and fees from Chinese listings

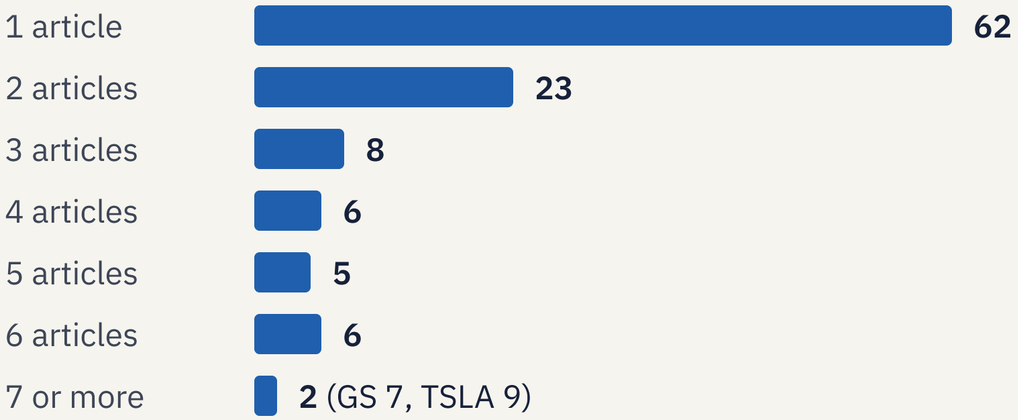
QCOM +0.74 · licence with Apple renewed (one article)

PYPL +0.72, HUM +0.71 · takeover rumours; Barclays upgrade (one article each)

Pattern: a few recent, company-specific events drive the extremes: downgrades and upgrades, guidance, a contract or licence. Only Nike is consistently negative across several weeks.

Coverage is wide but thin

Tickers by number of articles that mention them



112 of 500 companies appear (22 %); none from Materials.

62 tickers appear in a single article; the 10 most cited hold only 27 % of pairs.

Half the pairs are neutral (117 of 233): banks issuing ratings, media sources, comparisons.

Financials are cited most (65 pairs), mostly as neutral actors; **Consumer Discretionary** carries the most directional labels (30).

11 articles mention no S&P 500 company (Snap, GameStop, Rivian, BYD…).

What limits confidence in the numbers

One article per score

58 of the 80 non-zero scores rest on a single directional article; BX, BA, QCOM, PYPL and HUM are among the extremes.

Recency dominates

An article of 28 Sep weighs 0.97, one of 25 Aug 0.03 (30×). The 32 articles of the last week carry 63 % of the weighted signal.

Sums, not averages

More coverage means a larger score: Nike's −1.39 reflects four articles, not a stronger single event.

Label uncertainty

7 of 114 scores differ between passes (Alphabet +0.61 or +1.19); the five lowest are identical in all three. Accuracy ≈ 0.93, not 1.

100 articles over five weeks describe a news flow, not company value: the scores rank news tone, they are not investment signals.

Optional challenges: data, prompt, cost

Data quality

Zero-width spaces split "Tesla" in N027 and N100: removed on a copy.

Cyrillic look-alikes "Tesla" in N003 and N098: recognised by the model, flagged by code.

3 dual share classes merged; brands (WSJ, CNN, iShares, Xbox) mapped to parents; "Target", "Block", social network "X" blocked as false tickers.

Prompt iteration

v2 paragraph inventory: analyst banks recovered.

v4 structured prompt + examples: Waymo → Alphabet found.

v5 comparisons neutral: peers fixed, sources lost.

v6 roles enforced by the schema: 0.907 → 0.92 per pass.

Cost

One cold pass: $0.030 per article, 86 % of it writing the shared 14k-token prefix to the provider cache.

Passes 2–3 reuse that prefix: about $0.006 per article.

Re-reading only flagged articles: 66 % cheaper than reviewing all.

Assumptions, limits and next steps

ASSUMPTIONS

Labels follow the PDF definitions: implication for the company at publication time, judged from the article alone.

Brand → parent links come from established knowledge, not from the CSV.

Dates are New York local time; an article after 29 Sep would be rejected, not truncated.

NEXT STEPS

Flag look-alike Unicode in the input audit, not only invisible spaces.

Reuse the cached prompt prefix across runs to cut the first pass's cost.

github.com/sonalain077/Whitehole_project · **python -m aie2609 run --max-cost-usd 10**